Health Care AI

Software Requirements Specification (SRS)

SENG 384

CodeIsUnbreakable

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Software Requirements Specification (SRS)..... 1
1. Introduction..... 1
1.1 Purpose..... 1
1.2 Scope..... 2
1.3 Definitions and Abbreviations..... 2
1.4 Intended Audience..... 2
2. Overall Description..... 2
2.1 Product Perspective..... 2
2.2 User Roles..... 2
2.3 Assumptions and Dependencies..... 3
Assumptions..... 3
Dependencies..... 3
2.4 Constraints..... 4
3. Functional Requirements..... 4
3.1 User Registration & Access Control..... 4
3.2 Post Management..... 5
3.3 Search & Matching..... 5
3.4 Meeting Request Workflow..... 6
3.5 Administrative Dashboard..... 6
3.6 Activity Logging & Audit Trail..... 7
4. Non-Functional Requirements..... 7
5. Use Cases..... 9
5.2 UC-02: User Requests a Meeting (Inbox Workflow)..... 10
5.3 UC-03: Post Owner Marks "Partner Found"..... 10
6. Data Model..... 11
7. Interface Requirements..... 11
7.1 User Interface (UI)..... 11
7.2 External System Interfaces..... 12
8. Requirements Traceability Matrix..... 12
9. Appendices..... 12
9.1 Use Case Diagram..... 12
9.2 ER Diagram..... 13
9.3 Wireframes..... 15

1. Introduction

1.1 Purpose

The purpose of this document is to specify the complete functional and non-functional requirements for the **HEALTH AI Co-Creation & Innovation Platform**. This SRS is being prepared to guide the development team in building a secure, structured matchmaking environment for interdisciplinary healthcare technology innovation.

1.2 Scope

The platform enables structured partner discovery between healthcare professionals and engineers. It provides a secure matchmaking infrastructure and meeting initiation facilitator while intentionally excluding financial transactions, contract management, medical advice, and technical document storage.

1.3 Definitions and Abbreviations

- **SRS**: Software Requirements Specification.
- **RBAC**: Role-Based Access Control.
- **GDPR**: General Data Protection Regulation.
- **NDA**: Non-Disclosure Agreement.
- **WCAG**: Web Content Accessibility Guidelines.

1.4 Intended Audience

This document is intended for the development team, the course instructor, and stakeholders of the Erasmus+ HEALTH-AI Project who are evaluating the system for usability and stability.

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2. Overall Description

2.1 Product Perspective

Currently, multidisciplinary health-tech innovation relies on coincidence or inefficient personal networks, which slows the transformation of clinical ideas into viable engineering solutions. This platform addresses that gap by providing a secure, web-based infrastructure for structured access to complementary expertise.

2.2 User Roles

Role	Description	Key Permissions
Engineer	Technical professional seeking clinical domain knowledge and validation pathways.	Create/edit posts, browse healthcare projects, request meetings. respond to requests.
Healthcare Professional	Medical expert with innovation ideas lacking technical infrastructure for implementation.	Create/edit posts, browse engineering expertise, request meetings. respond to requests.
Admin	System moderator responsible for platform integrity and compliance.	Manage users and posts, remove inappropriate content, view and extract system logs.

2.3 Assumptions and Dependencies

Assumptions

- **Email Verification**: It is assumed all users have access to a valid institutional **.edu** email address for mandatory registration.
- **External Meetings**: It is assumed that users will conduct meetings via external platforms (e.g., Zoom, Microsoft Teams), as the system only facilitates initial contact and scheduling.
- **Connectivity**: Users are assumed to have a stable internet connection and modern web browser support.
- **Browser Compatibility:** Users access the system through modern web browsers.

Dependencies

- **Email Services:** Required for account verification and notifications.
- **Authentication Infrastructure:** Secure mechanisms for user authentication and session management.
- **Hosting & Infrastructure:** Reliable cloud or server infrastructure to ensure system availability and performance.

2.4 Constraints

- **Data Minimization**: No patient data, confidential technical documentation, or intellectual property files shall be stored on the platform.
- **Legal Compliance**: The system must be GDPR-compliant and enforce NDA acceptance before sharing meeting details.
- **Access Restriction**: Registration is strictly limited to institutional emails; personal providers are not permitted.
- **Scope Limitation**: No financial transactions or contract management will be performed within the system.
- **Predefined Roles:** Users must select predefined roles (Engineer / Healthcare Professional / Admin).
- **Security Constraints:** All communication must occur over **HTTPS**, Passwords must be securely stored using encryption/hashing, Session management must include timeout and protection against unauthorized access.

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3. Functional Requirements

3.1 User Registration & Access Control

ID	Requirement Description	Priority	Source
FR-01	The system shall only allow registration with institutional .edu email addresses.	High	Brief 4.1
FR-02	The system shall implement an email verification mechanism required for account activation.	High	Brief 4.1
FR-03	The user shall select a role (Engineer / Healthcare Professional / Admin) during registration.	High	Brief 4.1
FR-04	The system shall enforce Role-Based Access Control (RBAC) for all platform features.	High	Brief 4.1

FR-05	The system shall allow users to permanently delete their account and associated data (GDPR Right to Erasure).	Medium	Brief 5.1
FR-06	The system shall provide a function for users to export their profile info, posts, and meeting history in a downloadable format.	Low	Brief 5.1

3.2 Post Management

ID	Requirement Description	Priority	Source
FR-10	The "Create Post" form must include: Title, Working Domain, Required Expertise, Project Stage, Confidentiality Level, City, and Description.	High	User Guide 3.2
FR-11	The system shall enforce a mandatory post lifecycle: Draft, Active, Meeting Scheduled, Partner Found, Expired.	High	Brief 4.2
FR-12	Users shall be able to mark a post as "Partner Found" to close it from public discovery.	Medium	Brief 4.2
FR-13	The system shall prevent the upload of technical documents or patient data files.	High	Brief 4.2
FR-14	Users shall specify a "Confidentiality Level" (Public pitch vs. Details in meeting only) for each post.	Medium	Brief 4.2
FR-14	Users can archive posts		

3.3 Search & Matching

ID	Requirement Description	Priority	Source
FR-20	The system shall allow filtering posts by Domain, Required Expertise, City, Country, and Project Stage.	High	Brief 4.3

FR-21	The system shall highlight city-based matches to encourage local collaboration.	Medium	Brief 4.3
FR-22	The system shall provide a match explanation based on shared expertise tags.	Low	Brief 4.3

3.4 Meeting Request Workflow

ID	Requirement Description	Priority	Source
FR-30	The system shall provide a **Structured Inbox** to manage "Expressions of Interest" and short messages.	High	Brief 4.4
FR-31	The system shall mandate NDA acceptance before allowing a user to send or accept a formal Meeting Request.	High	Brief 4.4
FR-32	The workflow shall include a step for proposing and accepting specific date/time slots.	High	Brief 4.4
FR-33	The system shall allow users to share external meeting links (Zoom/Teams) only after mutual confirmation.	High	Brief 4.4

3.5 Administrative Dashboard

ID	Requirement Description	Priority	Source
FR-40	Admins shall have the ability to view all posts and remove content deemed inappropriate.	High	Brief 4.5

FR-41	Admins shall be able to suspend or deactivate user accounts for policy violations.	High	Brief 4.5
FR-42	The dashboard shall display platform statistics including total posts, matches, and user role distribution.	Medium	Brief 4.5
FR-43	Administrators shall be able to extract logs as CSV files	Medium	Brief 4.6

3.6 Activity Logging & Audit Trail

ID	Requirement Description	Priority	Source
FR-50	The system shall log all security events, including logins, failed attempts, and admin actions.	High	Brief 4.6
FR-51	Logs must record Timestamp, User ID, Action Type, and Result Status without containing patient data.	High	Brief 4.6
FR-52	Logs shall be tamper-resistant and retained for a defined period of 24 months.	Medium	Brief 4.6

Following our updated requirements and the structured workflow from the User Guide, here is the completion of your SRS starting from **Section 4: Non-Functional Requirements**.

4. Non-Functional Requirements

These requirements define the quality attributes and constraints of the system using measurable targets based on sections 5.1–5.4 of the project brief.

ID	Requirement	Metric / Target	Category
NFR-01	Search results shall be returned within a specified time.	< 1.5 seconds	Performance
NFR-02	Page load time shall be limited.	< 3 seconds	Performance
NFR-03	The system shall support a high volume of active users simultaneously.	1,000 concurrent users	Performance
NFR-04	Users must understand the platform's core purpose quickly upon landing.	< 20 seconds	Usability
NFR-05	The process for creating a new announcement must be efficient.	< 3 minutes	Usability
NFR-06	Passwords must be stored using industry-standard hashing.	Encrypted storage	Security
NFR-07	The platform must ensure accessibility for users with disabilities.	WCAG 2.1 Compliance	Accessibility
NFR-08	Data minimization must be enforced to protect user privacy.	No patient/technical data storage	GDPR/Privacy
NFR-09	All communication shall be secured.	HTTPS mandatory	Security

NFR-10	The system shall ensure secure session management.	Token-based authentication	Security
NFR-11	The system shall use clear and simple language.	Plain language UI	Usability
NFR-12	System status shall be clearly visible to users.	Clear status indicators	Usability

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5. Use Cases

These use cases describe the core interactions within the platform based on the functional workflows.

Name	**UC-01: User Creates a Post**
Actor(s)	Engineer or Healthcare Professional
Precondition	The user must be logged in with a verified account.
Main Flow	1. The user clicks the “New Post” button on the Dashboard. 2. The user fills in the mandatory fields: Title, Working Domain, Required Expertise, Project Stage, Confidentiality Level, City, and Description. 3. The user selects whether to save as a **Draft** or **Publish** immediately. 4. The system validates the input and saves the post.
Postcondition	The post is saved with the status “Active” (if published) or “Draft”.
Alternative	**2a.** If a required field is left blank, the system displays an error message .

Flow	**2b.** If the user attempts to upload a file, the system blocks the action.

5.2 UC-02: User Requests a Meeting (Inbox Workflow)

Name	**UC-02: User Requests a Meeting**
Actor(s)	Engineer or Healthcare Professional
Precondition	The actor must be logged in and viewing an "Active" post.
Main Flow	1. The actor clicks "Express Interest" and sends an optional short message. 2. The post owner receives a notification in their **Inbox**. 3. The post owner acknowledges interest and proposes specific date/time slots. 4. The actor selects a slot and **accepts the NDA**.
Postcondition	Both parties receive a confirmation; the meeting is scheduled.
Alternative Flow	4a. If the actor declines the NDA, the meeting request is terminated.

5.3 UC-03: Post Owner Marks "Partner Found"

Name	**UC-03: Marking Partner Found**
Actor(s)	Post Owner (Doctor or Engineer)
Precondition	The post must have a "Meeting Scheduled" or "Active" status.

Main Flow	1. The owner opens their specific post detail page. 2. The owner clicks the “Mark as Partner Found” button. 3. The owner confirms the action in a pop-up dialog.
Postcondition	Post status changes to “Partner Found” (Closed) and is hidden from search.
Alternative Flow	3a. If the owner cancels the confirmation, the post remains in its current status.

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6. Data Model

The following table outlines the core data entities and their relationships required for the matchmaking infrastructure.

Entity	Key Fields	Relationships
User	UserID, .edu Email, Password, Role, City, Institution	One User can own many Posts; can send/receive many MeetingRequests.
Post	PostID, Title, Domain, Stage, Confidentiality, City, Status	Belong to one User; are the target of many MeetingRequests.
MeetingRequest	RequestID, TimeSlots, NDA_Accepted (Boolean), Status	Connects a Requester to a Post and its Owner.
ActivityLog	LogID, Timestamp, UserID, ActionType, ResultStatus	Linked to a User and an Action for audit purposes.

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7. Interface Requirements

7.1 User Interface (UI)

The platform will consist of a clean, professional web interface with the following main screens:

- **Landing Page/Registration:** Highlighting the platform purpose within 20 seconds.
- **User Dashboard:** A central feed of active posts with a notification indicator for the **Inbox**.
- **Structured Inbox:** The primary area for managing interest messages, proposing time slots, and accepting NDAs.
- **Admin Panel:** Restricted view for user suspension and post moderation.

7.2 External System Interfaces

- **Email Service:** Used for mandatory .edu email verification and meeting notifications.
- **External Meeting Links:** The system will store text strings for Zoom/Teams links, but actual video/audio interactions are handled by these external providers.

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8. Requirements Traceability Matrix

This matrix ensures every functional requirement is mapped to its source and a testing scenario.

Req. ID	Requirement Summary	Source (Brief)	Related Use Case
FR-01	.edu email restriction	Brief 4.1	Registration Flow
FR-10	Post creation fields	Brief 4.2	UC-01
FR-32	Mandatory NDA acceptance	Brief 4.4	UC-02
FR-12	Mark partner found	Brief 4.2	UC-03
FR-40	Admin post removal	Brief 4.5	Admin Flow

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9. Appendices

9.1 Use Case Diagram

Visual representation of actors (Engineer, Doctor, Admin) interacting with the platform.

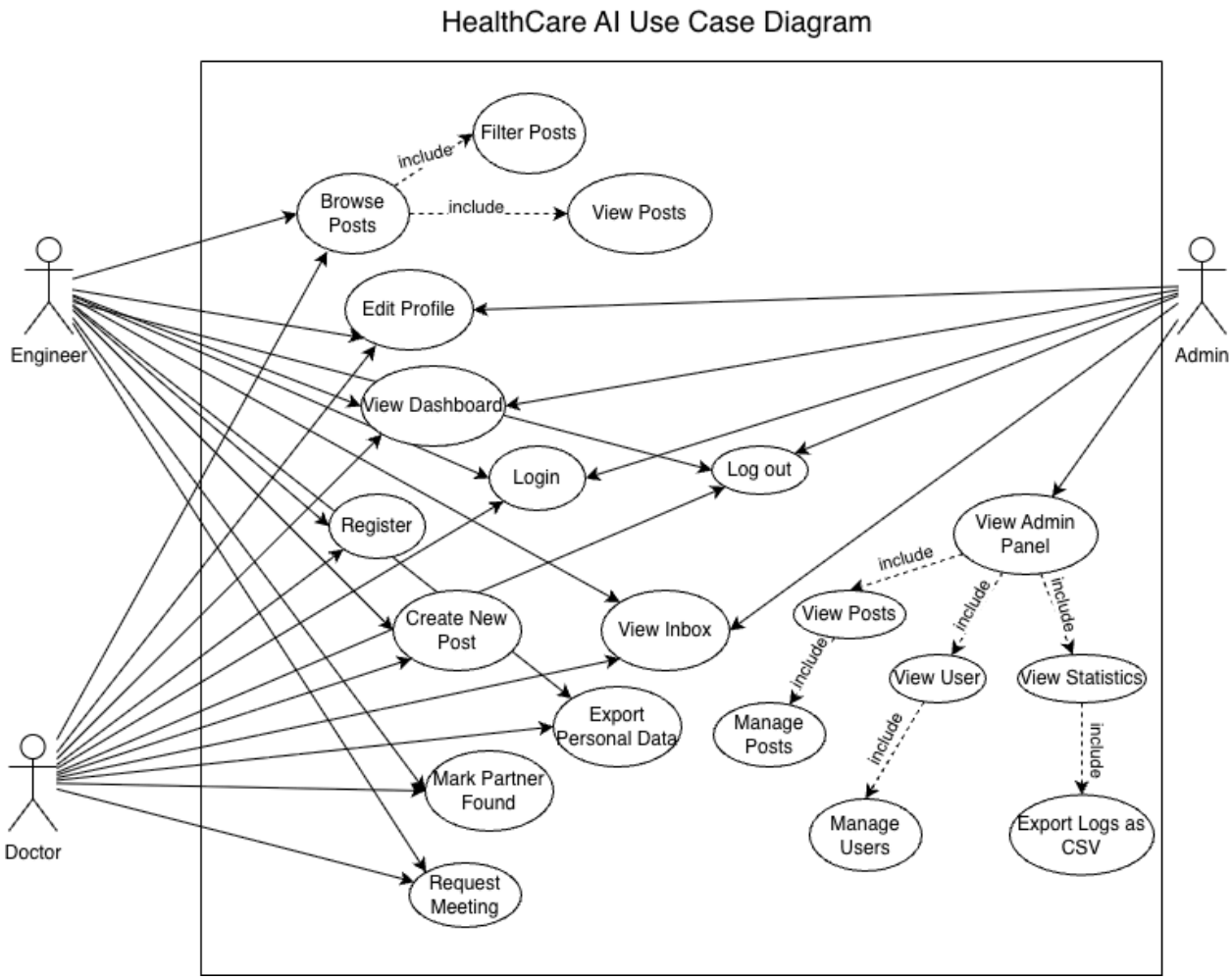

Image 1: Use-Case Diagram

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9.2 ER Diagram

A diagram showing the one-to-many relationships between Users, Posts, and Logs.

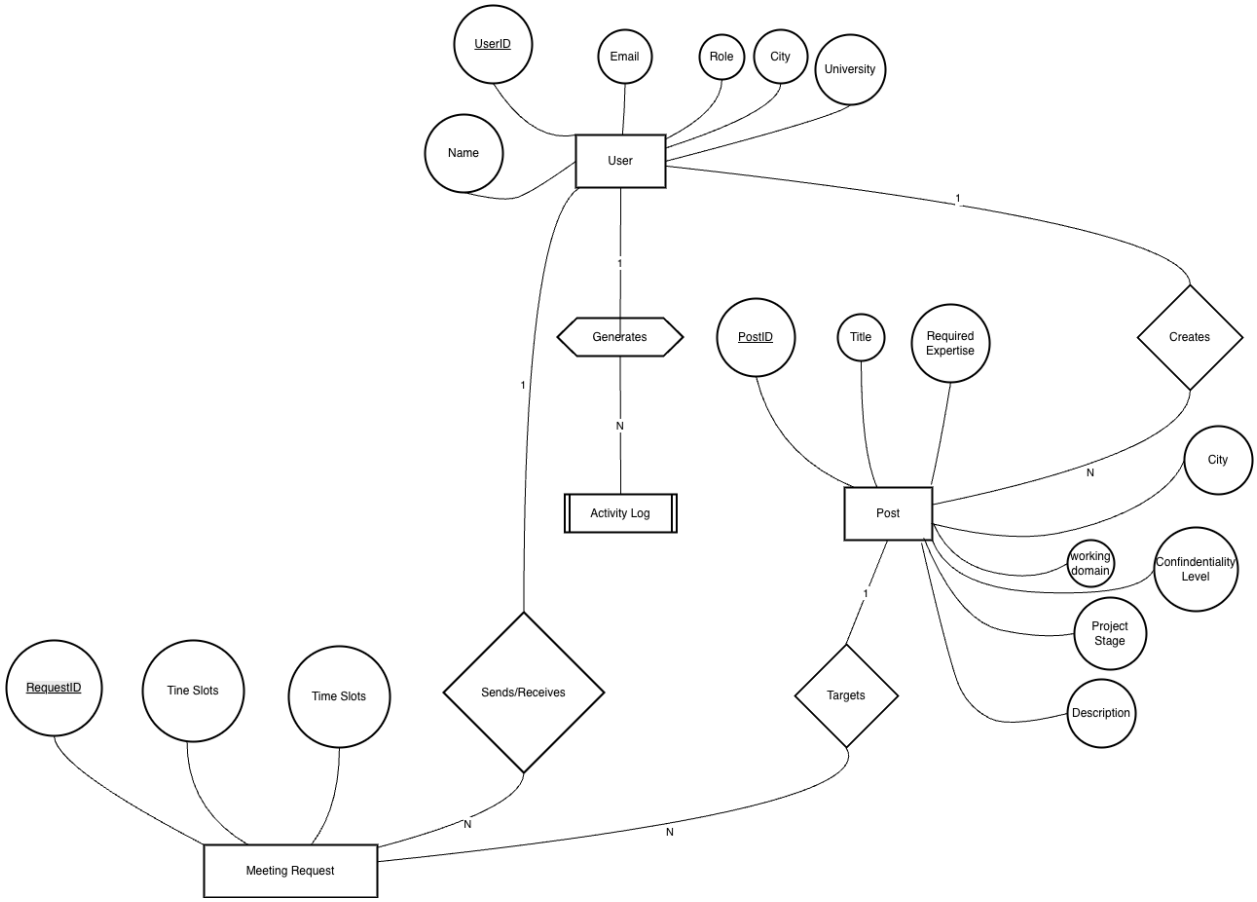

Image 2: ER Diagram

9.3 Wireframes

Mockups of the Dashboard and the Structured Inbox.

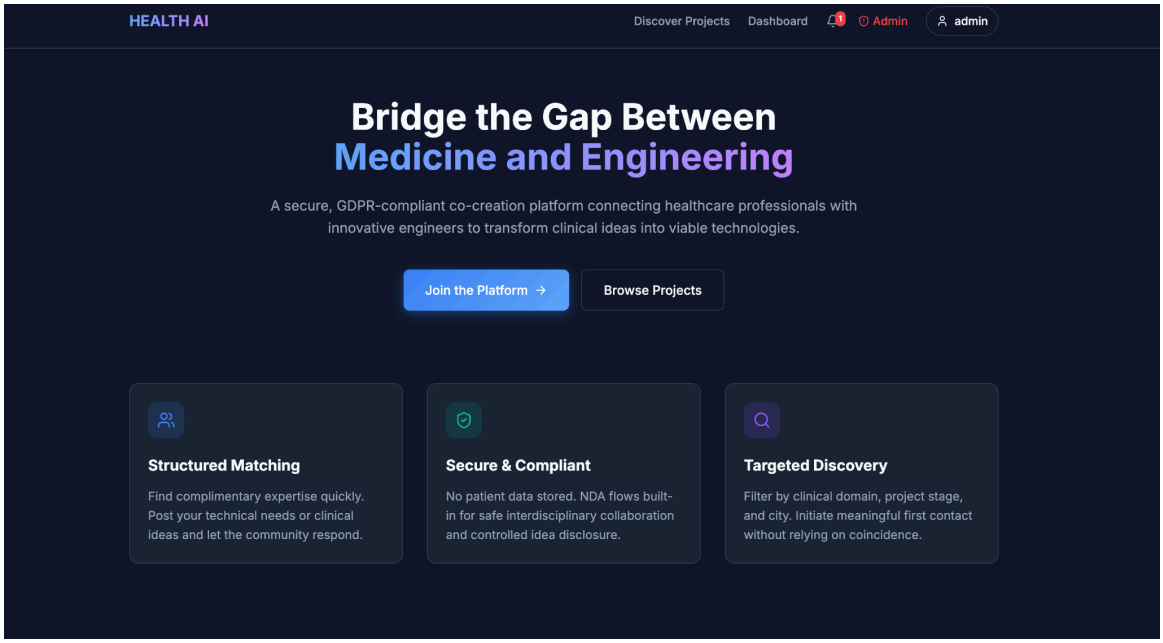

Image 3: Main Page

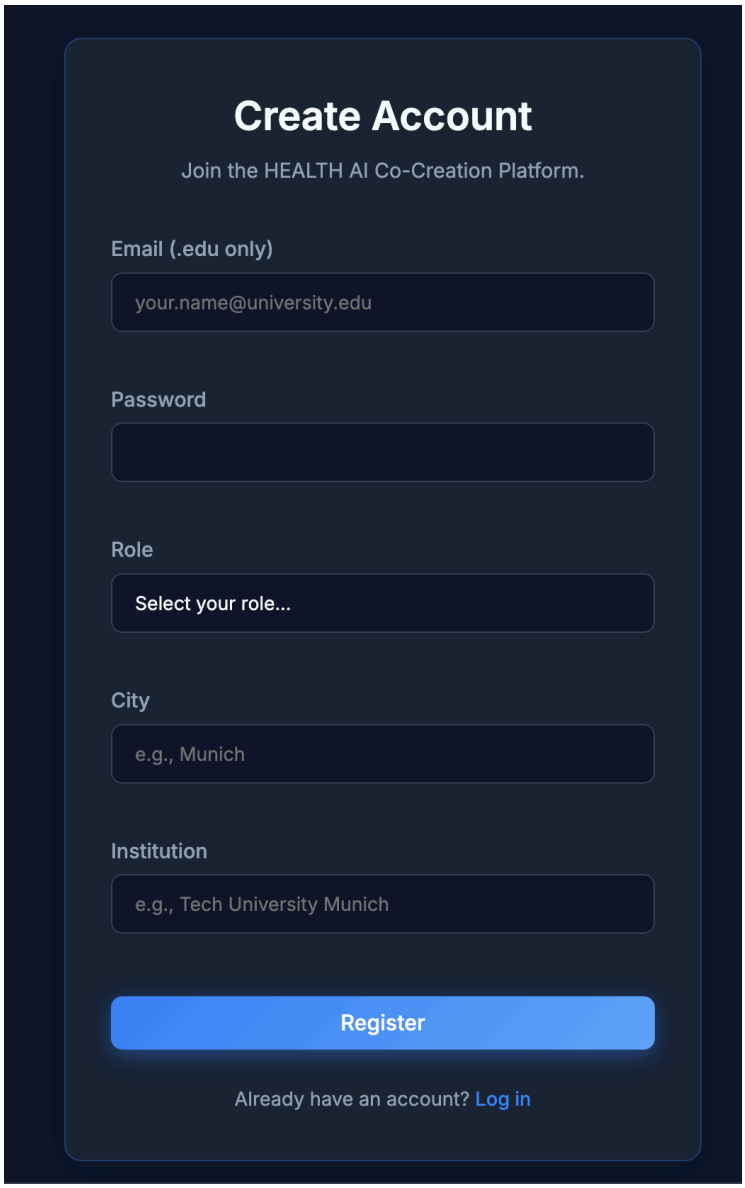

Image 4: Create Account Page

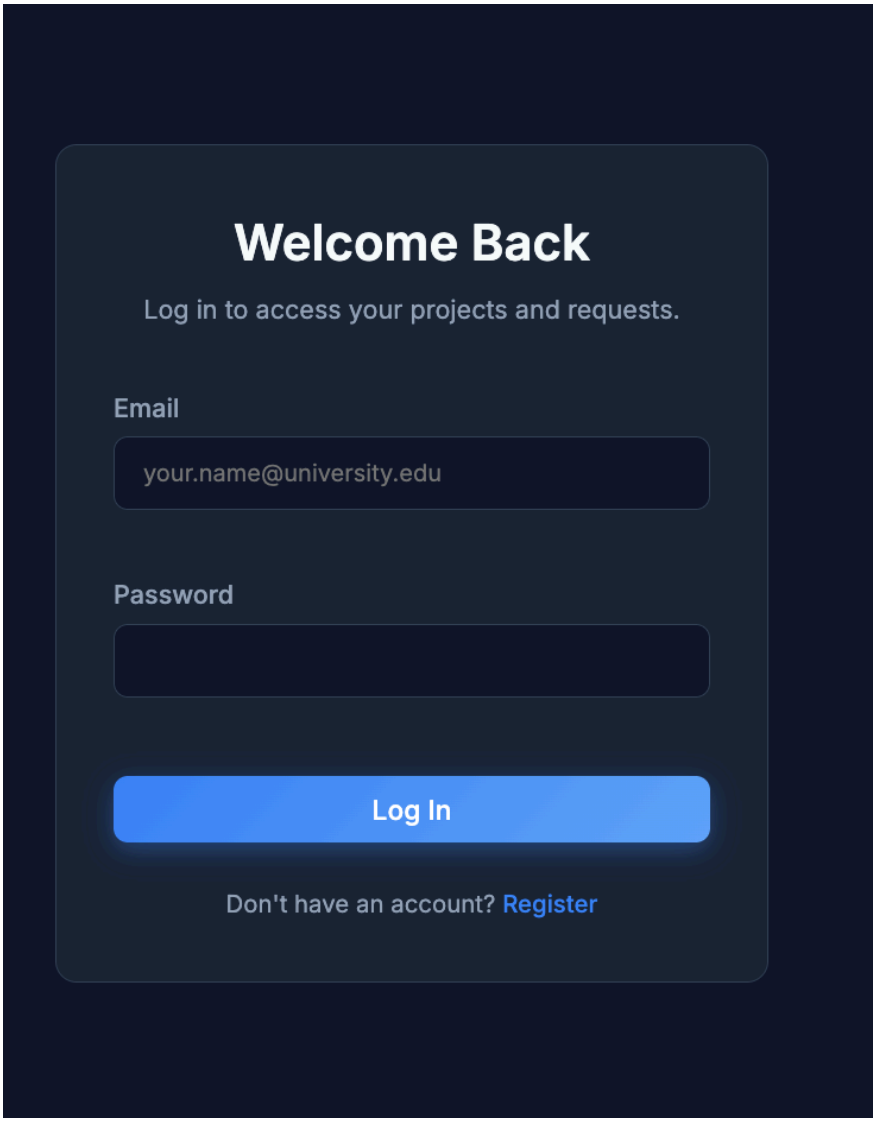

Image 5: Log In Page

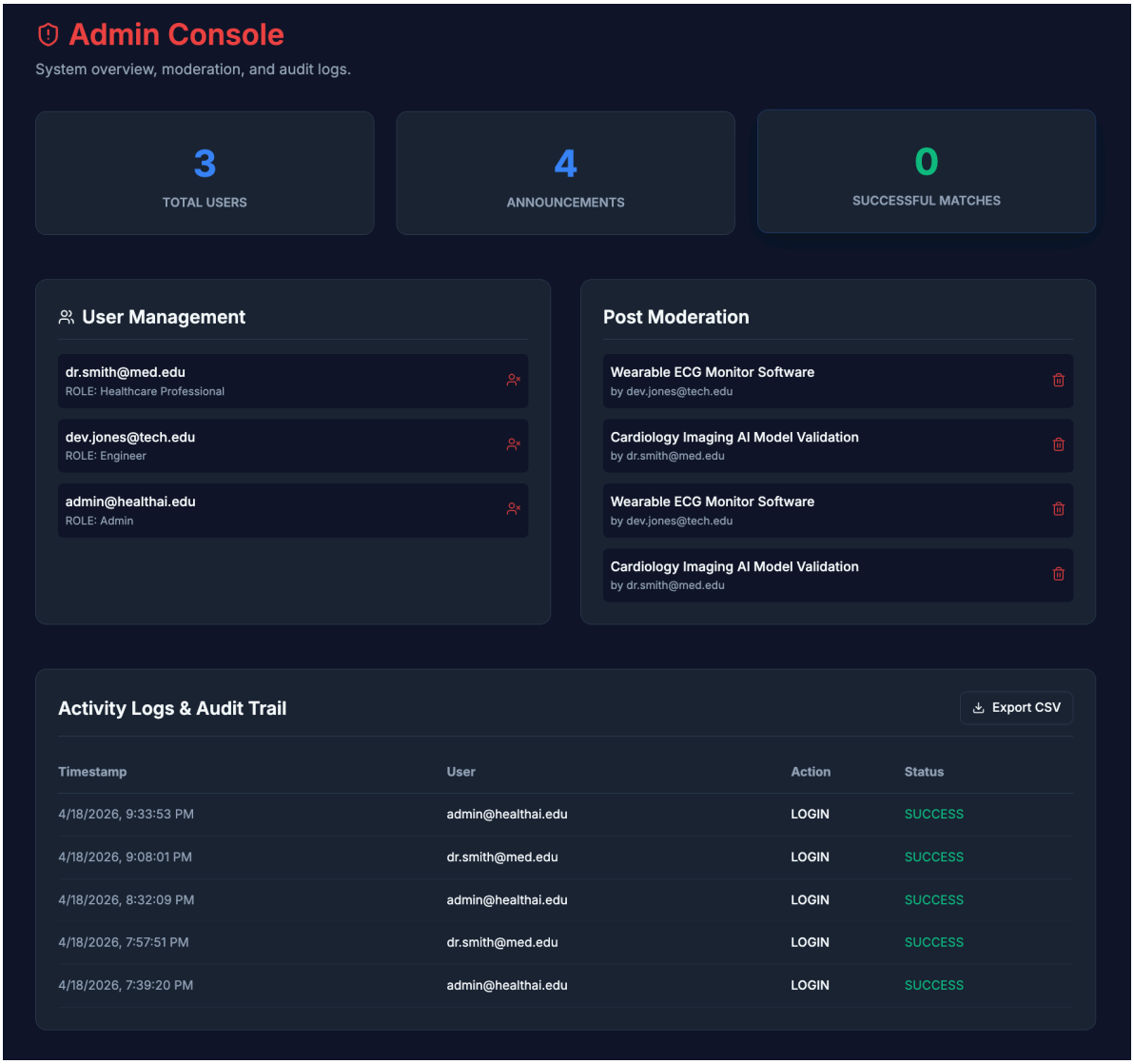

Image 6: Admin Page

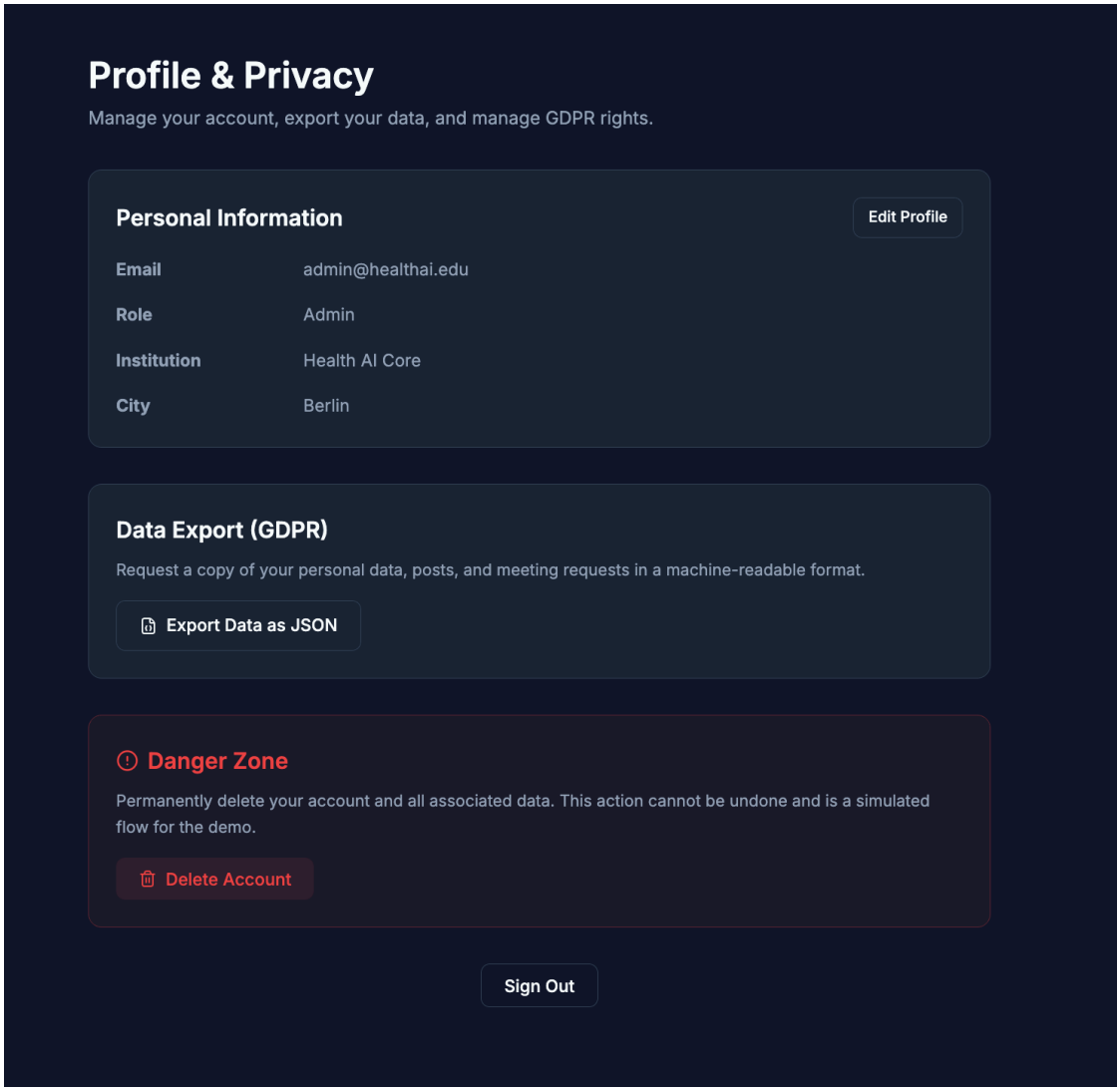

Image 7: Profile Page

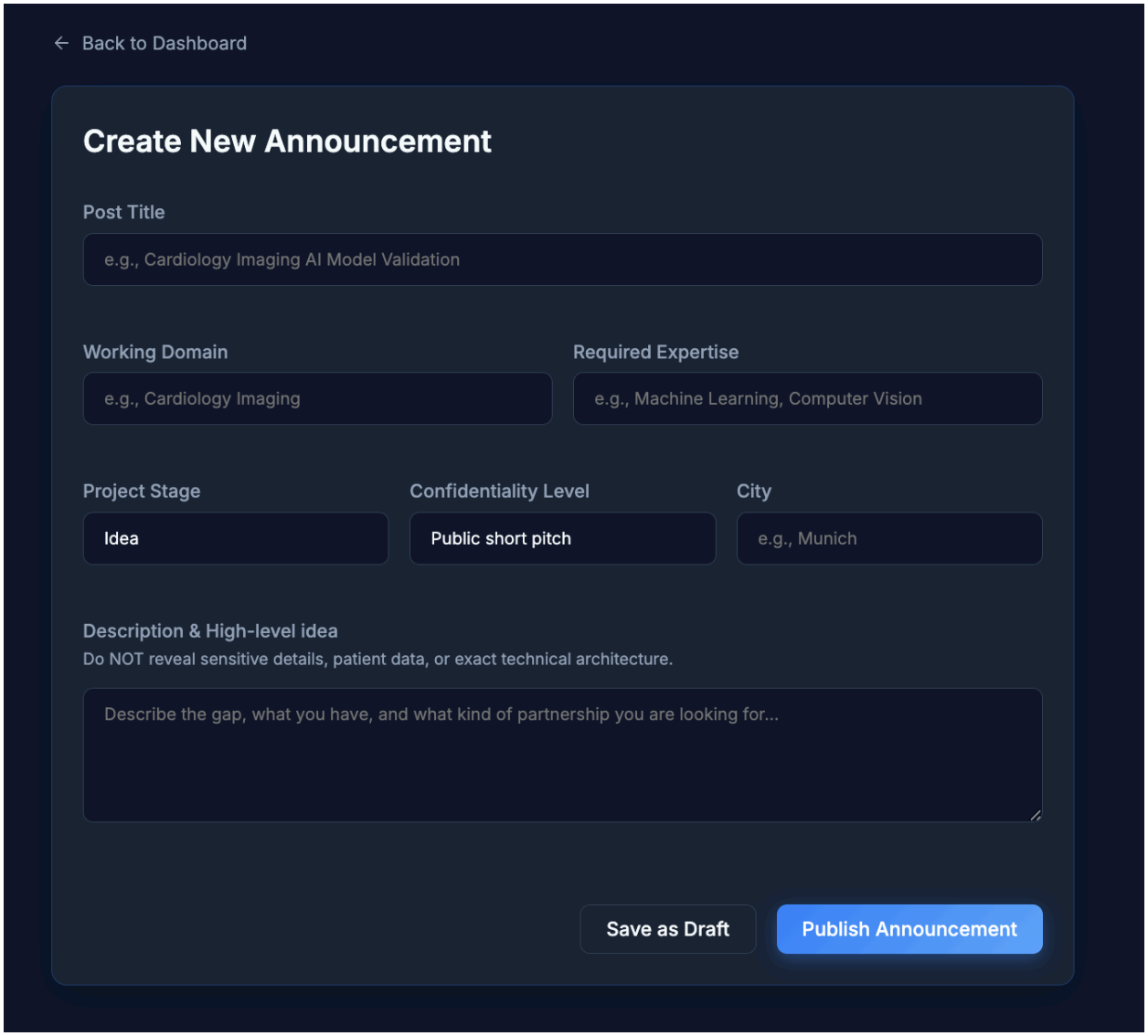

Image 8: Create Post Page

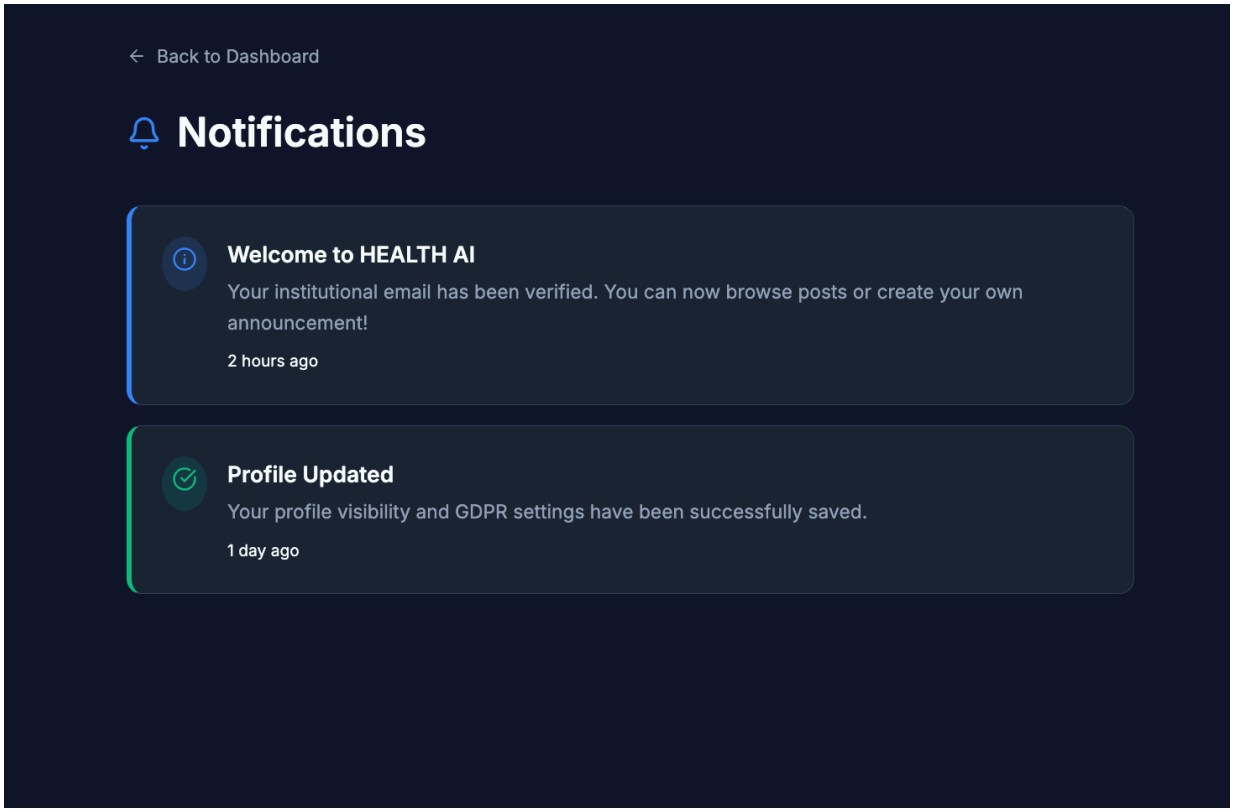

Image 9: Notifications Page

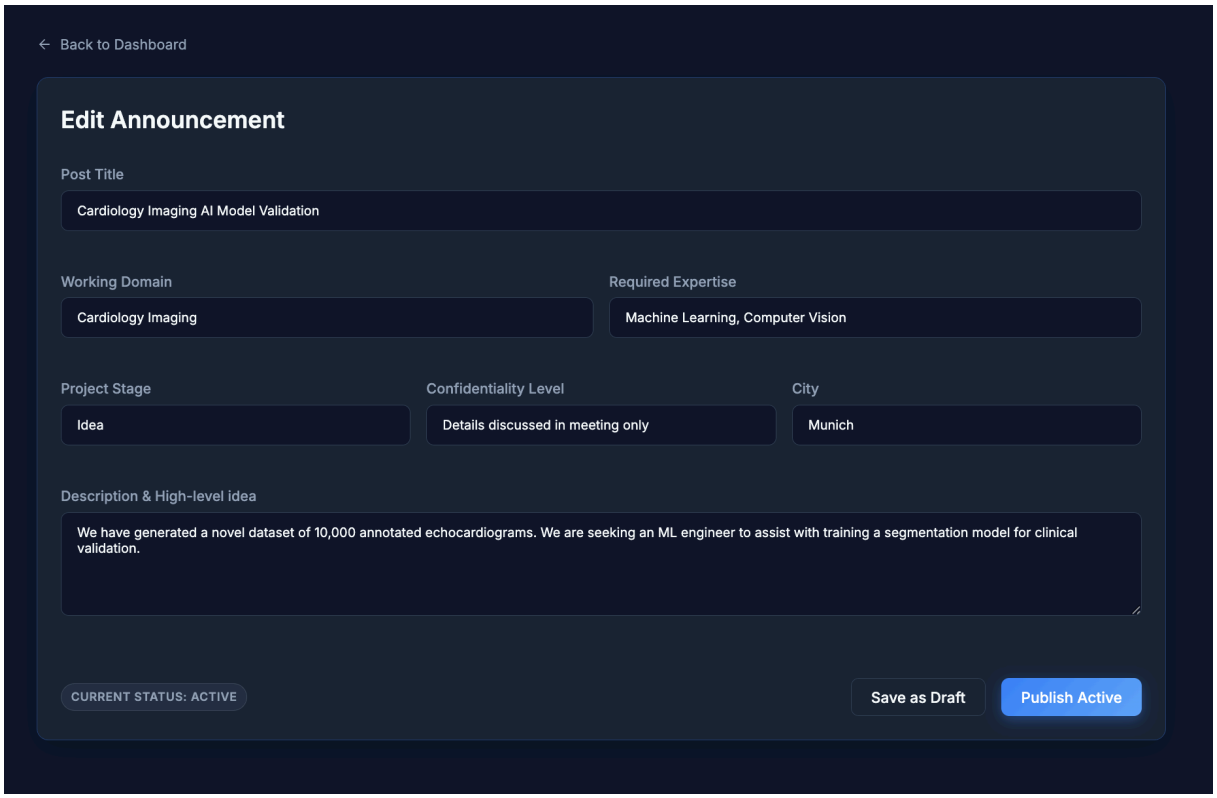

Image 10: Edit Post Page

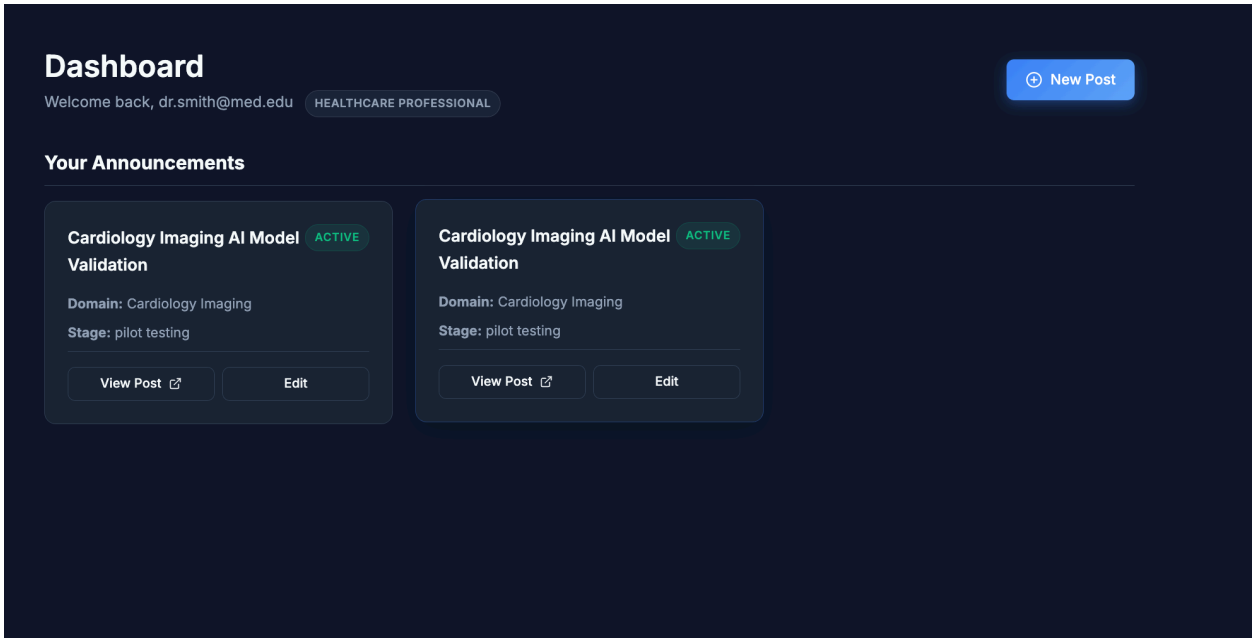

Image 11: Dashboard Page

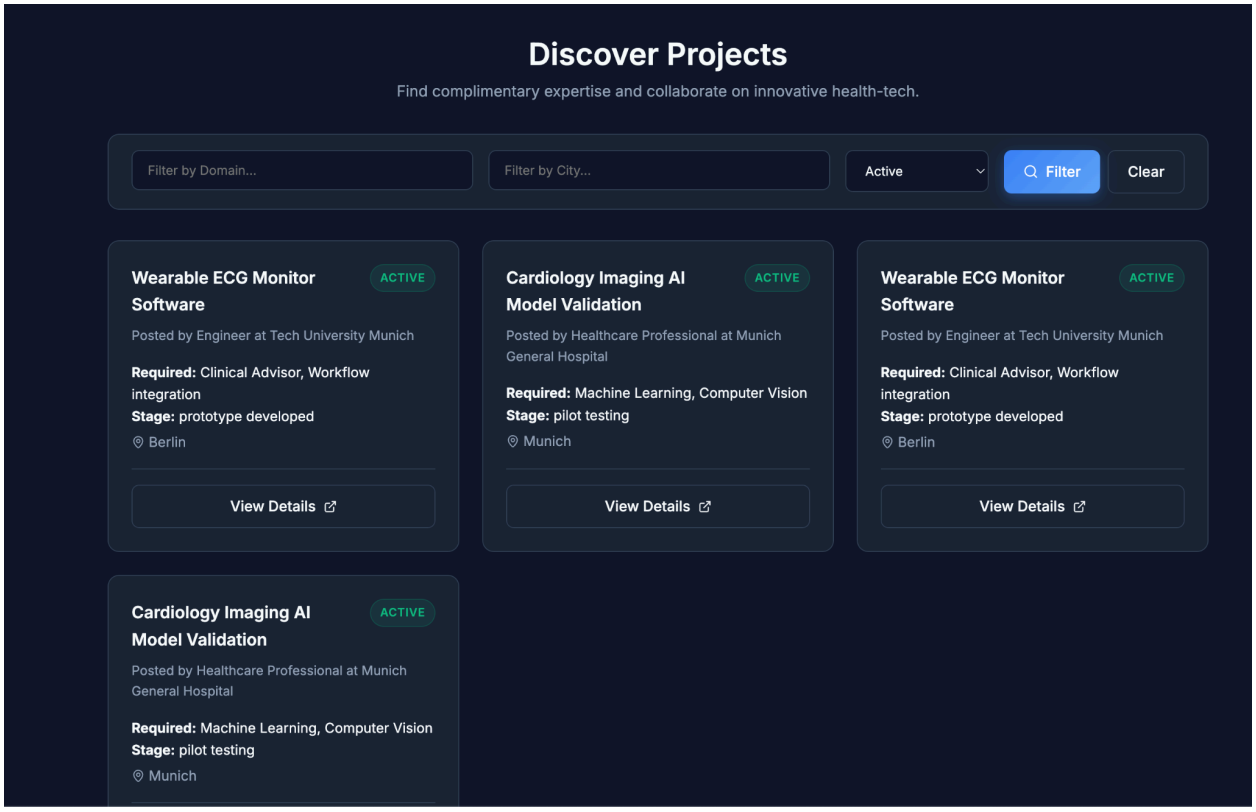

Image 12: Search Post Page